Thus, the full Lagrangian for the witness particle (relativistic charged particle in electromagnetic potentials ϕ and A) is:

$$L = -mc^2\sqrt{1-v^2/c^2} + q\vec{A}\cdot\vec{v} - q\phi, \quad (I)$$

such that:

$$S = \int L\,dt = \underbrace{-mc^2\int \frac{dt}{\gamma}}_{S_{rel}} + \underbrace{q\int(\vec{A}\cdot\vec{v} - \phi)\,dt}_{S_{Lor.}}$$

- $dS_{rel} = -mc^2 d\tau$

  ~ a particle with mass accumulates action proportional to its proper time $d\tau$

- $dS_{Lor} = q(\vec{A}\cdot d\vec{r} - \phi dt)$

  ~ a charged particle accumulates action by moving through electromagnetic potentials

where $\vec{A}$ and ϕ are the electromag. potentials defined by the ME for the source particle together (which gives $\vec{E}$ and $\vec{B}$) with the definitions $\vec{E} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla}\times\vec{A}$.

Then, applying the Euler-Lagrange equations to (I):

$$\frac{d}{dt}\left(\frac{\partial L}{\partial\vec{v}}\right) - \vec{\nabla}L = \frac{d}{dt}\left(\cancel{-}mc^2\frac{\cancel{-2}\vec{v}/c^2}{\cancel{2}\sqrt{1-v^2/c^2}} + q\vec{A}\right) - q\vec{\nabla}\left(\vec{A}\cdot\vec{v} - \phi\right) = \frac{d}{dt}\left(\gamma m\vec{v} + q\vec{A}\right) - q\vec{\nabla}\left(A_s v_s - \phi\right)$$

$v_x, v_y \ll v_s$

$$\Rightarrow \frac{d}{dt}\left(\vec{p}_{rel} + q\vec{A}\right) = q\vec{\nabla}\left(A_s v_s - \phi\right)$$

Integrating both sides:

$$\left[\vec{p} + \cancel{q\vec{A}}^{\,0}\right]_{t=-\infty}^{t=+\infty} = \underbrace{\Delta\vec{p}}_{\text{momentum change}} = q\cdot\int_{-\infty}^{\infty}\vec{\nabla}(A_s v_s - \phi)\,dt = q\vec{\nabla}\int_{-\infty}^{\infty}(A_s v_s - \phi)\,dt$$

assumption that x, y are fixed with t

gradient of a scalar function

as we desired!

Remembering that $\vec{W} = \frac{c}{qQ}(\Delta p_x, \Delta p_y, -\Delta p_z)$, and seeing that we can write $\vec{\Delta p} = \frac{qQ}{c}\vec{\nabla}_R W$, where $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial s}\right) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial(-z)}\right)$

$\vec{R} = (x_w, y_w, -z)$ ($z \leftrightarrows s$), $\Rightarrow \vec{\Delta p} = \frac{qQ}{c}\left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z}\right)$, we have:

$$\Rightarrow (W_x, W_y, W_{\parallel}) = \frac{c}{\cancel{qQ}}\cdot\frac{\cancel{qQ}}{\cancel{c}}\left(\underbrace{\frac{\partial W}{\partial x_w}}_{\propto\Delta p_x}, \underbrace{\frac{\partial W}{\partial y_w}}_{\propto\Delta p_y}, \underbrace{\frac{\partial W}{\partial z}}_{\propto -\Delta p_s}\right) \Rightarrow \begin{cases} \vec{W_\perp} = \vec{\nabla}_{\perp,w} W \\ W_{\parallel} = \frac{\partial W}{\partial z} \end{cases}$$

!

So, in short: $\overset{\curvearrowright kg\cdot m/s}{\overrightarrow{\Delta p}} = \vec{\nabla}\underbrace{\left(q\int_{-\infty}^{\infty} (A_s v_s - \phi)\, dt \right)}_{f(\vec{r})} = \frac{q\cdot Q}{c}\left(w_x, w_y, -w_s \right)$ $\quad kg\, m^2/s^2 \cdot 1/C^2 = J/C^2 = V/C$

$$\Rightarrow \left(W_x, W_y, -W_{\parallel} \right) = \frac{c}{qQ}\cdot \vec{\nabla} f = \vec{\nabla}\left(\frac{c}{Q}\int_{-\infty}^{\infty} \overbrace{(A_s v_s - \phi)}^{J/C}\, dt \right) = \vec{\nabla}_R\, W(\vec{\rho_s}, \vec{\rho_w}, z) \quad (J/C^2)\cdot m$$

$$\left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial s} \right) \underset{s\sim -z}{=} \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial(-z)} \right)$$

And thus:

- $\left(W_x, W_y, -W_{\parallel} \right) = \vec{\nabla}_R W \Rightarrow \begin{cases} (W_x, W_y) = \overrightarrow{W_\perp} = \vec{\nabla}_{w,\perp} W \\ W_{\parallel} = -\frac{\partial}{\partial(-z)} W = \frac{\partial}{\partial z} W \end{cases}$

Furthermore, this potential formalism enables directly obtaining the so-called Panofsky-Wenzel theorem from the second cross derivatives of W:

$$\frac{\partial \overrightarrow{W_\perp}}{\partial z} = \frac{\partial}{\partial z}\left(\vec{\nabla}_{w,\perp} W \right) = \vec{\nabla}_{w,\perp}\left(\frac{\partial}{\partial z} W \right) = \vec{\nabla}_{w,\perp} W_{\parallel} \longrightarrow \frac{\partial \overrightarrow{W_\perp}}{\partial z} = \vec{\nabla}_{w,\perp} W_{\parallel} \qquad (\mathrm{II})$$

↳ This property of the wakes is very useful as it enables calculating $\overrightarrow{W_\perp}$ from $W_{\parallel}$ by a simple integration of (II):

$$\overrightarrow{W_\perp} = \int_{-\infty}^{z} \vec{\nabla}_{w,\perp} W_{\parallel}\, dz^*.$$

Furthermore, $W_{\parallel} = -\frac{c}{qQ}\overset{\text{impulse approx.}}{\Delta p_s} = -\frac{c}{qQ}\int_{-\infty}^{\infty} (F_L)_s\, dt = -\frac{c}{Q}\int_{-\infty}^{\infty} dt\, E_s\Big|_{s=ct-z}$ $\quad \vec{F}_L = q\left[E_s\hat{s} + (E_x - vB_y)\hat{x} + (E_y + vB_x)\hat{y} \right]$

$\Rightarrow$ All wake functions can be calculated from E_s!

$$W_{\parallel}(\vec{\rho_s}, \vec{\rho_w}, z) = -\frac{c}{Q}\int_{-\infty}^{\infty} dt\, E_s(\vec{\rho_s}, \vec{\rho_w}, z, t)\Big|_{s=ct-z} ; \quad \overrightarrow{W_\perp} = \int_{-\infty}^{z} \vec{\nabla}_{w,\perp} W_s\, dz^* \quad !$$

Another important property of the wake potential W is being an harmonic function in the transverse coordinates x and y $\left(\text{if } \beta = 1 \sim v = \beta c \right)^*$:

$$\nabla_\perp^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0 \qquad \sim \qquad \frac{\partial W_x}{\partial x} = -\frac{\partial W_y}{\partial y}$$

To prove this, we first remember that, from the ME in free space, the potentials ϕ and $\vec{A}$ satisfy the wave equation:

$$\nabla^2 \vec{A} - \overset{1/c^2}{\mu_0 \varepsilon_0} \frac{\partial^2 \vec{A}}{\partial t^2} = 0 \; ; \qquad \nabla^2 \phi - \mu_0 \varepsilon_0 \frac{\partial^2 \phi}{\partial t^2} = 0$$

Now, since $W = q \int_{-\infty}^{\infty} (A_s v - \phi)\, dt$, we have:

$$\overset{= \left(\nabla_\perp^2 + \frac{\partial^2}{\partial z^2}\right)}{\nabla^2} W = \nabla_\perp^2 W + \frac{\partial^2 W}{\partial z^2} = \nabla^2 \left(q \int_{-\infty}^{\infty} (A_s v_s - \phi)\, dt \right) = q \int_{-\infty}^{\infty} \left(c \nabla^2 A_s - \nabla^2 \phi \right) dt = q \cdot \int_{-\infty}^{\infty} \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \left(v A_s - \phi \right) dt$$

$$\Rightarrow \nabla_\perp^2 W = q \int_{-\infty}^{\infty} \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) \left(v A_s - \phi \right) dt \, .$$

Furthermore, $\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) = \left(\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right) \cdot \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right)$, and $z = \beta c t - s \Rightarrow \frac{d}{dt} = \left(\frac{\partial}{\partial t} + \frac{\partial z}{\partial t} \frac{\partial}{\partial z} \right) = \beta c \left(\frac{1}{\beta c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right)$, meaning that, for $\beta = 1$:

$$\nabla_\perp^2 W = q \int_{-\infty}^{\infty} c \frac{d}{dt} \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) \left(c A_s - \phi \right) dt = q c \cdot \left[\left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) \left(c A_s - \phi \right) \right]_{t=-\infty}^{t=\infty} = 0$$

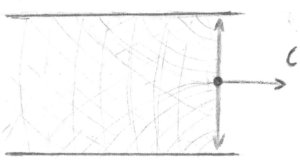
$\sim$ since $\vec{A}$ and ϕ vanish at $\pm\infty$

$$\therefore \quad \nabla_\perp^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0 \quad \blacksquare$$

1.5 - The "Causality" Principle

Related to this discussion is the remark that, in the ultra-relativistic limit $(v \to c)$, the wakes generated by a particle can only influence particles that are behind it *, since the EM fields (and thus the "wake wave-front") also propagate at speed c.

Wake analogy:



This is equivalent to saying that W must satisfy:

$$\boxed{W(\vec{\rho}_s, \vec{\rho}_w, z) = 0 \quad \forall \quad z < 0} \quad !$$

Which also implies: $\rightsquigarrow$ $\begin{cases} \bullet \ \vec{W_\perp} = \vec{\nabla}_{w,\perp} W = 0 & \forall \ z < 0 \\ \bullet \ W_\parallel = \dfrac{\partial W}{\partial z} = 0 & \forall \ z < 0 \end{cases}$

Even though this is a good approx. for real beams in SLS storage rings, the raw effective wakes from time-domain solvers do not respect it, as they are calculated from finite-length charge distributions

The wake formalism also applies to the modelling of other sources of col. effects, such as the interaction of the beam w/ ions in the vacuum chamber, and to model the effect of CSR on the beam ~ former → causal ✓ latter → not causal !